

VALUING AIR BARRIERS

Without national or local codes or regulations to mandate well-sealed apartment buildings, selling the cost benefit of tight building practices is key.



CMHC

Here is a piece of old news: Air leakage in apartment buildings is responsible for high energy bills, occupant complaints about drafts, and building envelope durability problems. Given that it is well known that apartment buildings tend to be leaky, the building performance community should be well on our way to eliminating the problem. In actual fact, we haven't progressed very far. What is surprising is that we know the types of problem that air leakage causes; we know how air leakage can be reduced with better design details, construction supervision, quality assurance, and testing—and yet it still doesn't happen.

Studies undertaken by the organization I belong to, the Canada Mortgage and Housing Corporation (CMHC), have found that air leakage in apartment buildings can contribute to as much as 20% of the annual space-heating energy load. It also represents a substantial proportion of the peak space-heating load (see Figure 1). Not surprisingly, air leakage control also represents a sub-

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stantial energy and cost savings opportunity, as highlighted in a couple of air leakage control projects monitored by CMHC in the 1990s (see "A Tale of Two Towers," p. 30).

How Bad Is It?

So how leaky are apartment buildings? This is not an easy question to answer, due to the way we measure the holes. The air leakage characteristics of large buildings, such as apartment buildings, are usually expressed in terms of the amount of air flow that can be induced through the building envelope at a certain indoor-outdoor pressure difference. Wind and the stack effect influence the indoor-outdoor pressure difference in tall buildings. Because of this, the pressure difference where the air leakage area is calculated must be well beyond the possible stack and wind pressures in order to make the tests valid. While

the measure of induced air flow at these elevated test pressures tells us nothing exact about the actual air leakage of the apartment building under normal conditions, it does provide a useful comparison against established benchmarks for airtightness and the results from other buildings. The Canadian unit of measurement for air leakage characteristics of large buildings used most often is liters per second per square meter of building envelope area, at 75 Pa indoor-outdoor pressure difference ($L/s/m^2$ at 75 Pa). The U.S. equivalent is expressed as cubic feet per minute per square foot at 0.3-inch WC (CFM/ft^2 at 0.3-in WC).

In January 2001, CMHC conducted a survey, undertaken by Proskiw Engineering, LTD, of Winnipeg, Manitoba, to gather air leakage data on large buildings. This work was done in order to establish baselines for air leakage rates and to assess whether there were any relationships

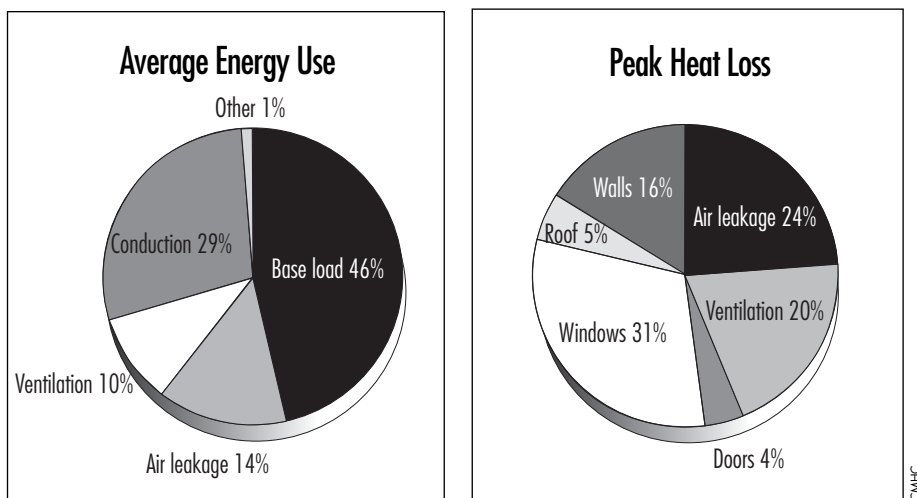


Figure 1. Air leakage results in a significant percentage of average annual energy use (14%) and peak heat loss (24%) in 10 apartment buildings.

between building age, size, wall construction, location, and use on the one hand, and the measured air leakage characteristics on the other. Table 1 summarizes the findings for the apartment buildings component of the project. The data in Table 1 are classified according to how the air leakage measurement was done. For Type 1 data, the whole building was tested and the total building

envelope area was used for normalizing the results. For Type 2 data, the whole building was tested but some other area (such as exterior wall area only) was used to normalize the air leakage flow induced at 75 Pa. For Type 3 data, the air leakage tests were performed within individual suites or floors and the air leakage induced at 75 Pa was normalized with respect to the exterior wall area

of the zone tested.

A review of the air leakage results that were obtained on different floors or different apartments within the same building found that the results could vary by a factor of 3. It appears that a significant variation in air barrier performance occurs over the envelope of a given building. Within the small data set of buildings, there was very little correlation between air leakage and wall construction type or building age. Newer buildings tended to be just as leaky as older buildings.

So how bad are the numbers? In Canada, the National Building Code contains recommendations that suggest that air leakage should be limited to 0.1 L/s/m² at 75 Pa (0.02 CFM/ft² at 0.3-in WC) for buildings with indoor humidity levels ranging from 27% to 55%—the range into which most multifamily buildings fall. A quick comparison shows that most apartment buildings are 30 to 40 times more leaky than the recommended limits.

What's more, many large-building green or energy efficiency programs—under which many advanced buildings have been

A Tale of Two Towers: Case Studies in Air Leakage Control

In 1990, CMHC, in cooperation with Ontario Hydro, commissioned a project to evaluate the extent of energy and cost savings possible through air leakage control in large multiunit residential buildings. The ambitious study evaluated the air leakage and energy consumption characteristics of two buildings before and after air sealing. The cost of the air sealing projects and the annual energy savings were used to determine the payback period.

The buildings, at the time, were 20 years old. The first, located in Ottawa, was 21 stories and 240 units. The second, located in Toronto, was 10 stories and 95 units. Both buildings were electrically heated. The results of the air sealing program are shown in Table A.

The air leakage program consisted primarily of sealing the tops and bottoms of the buildings. Garbage chutes and shafts, stairwell doors, penthouse machinery rooms, elevator shafts, and plumbing penetrations were sealed at the roof level. Wiring penetrations behind electric baseboard heaters and windows were sealed in the apartments.

The study demonstrated that air leakage control projects produced

tangible results and a very attractive return on investment for property owners. However, the "success" of the project begs the question, What if the buildings had been built better in the first place? Think of the energy and costs savings that would have accrued over the 20 years of operation! There is no doubt that these savings would have paid any incremental construction costs many times over.

Table A. Savings Due to Air Sealing

Building Location	Annual Energy Savings	Peak Load Reduction	Annual Cost Savings	Cost of Air Sealing	Simple Payback Period
Ottawa	164,870 kWh	85 kW	\$9,656	\$54,816	5.7 years
Toronto	63,340 kWh	42 kW	\$6,107	\$38,000	6.2 years



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Elevator cable penetrations are a significant leakage pathway.

constructed—have no concrete requirements for the design, specification, testing, and certification of air barrier performance. While the typical result is a well-insulated structure with the most innovative, high-efficiency mechanical and electrical systems, the integrity of the air barrier system remains unknown. It is simply assumed, with a hope and a prayer, that such buildings will actually achieve performance targets over the long term.

Now the Good News

There are now many building materials available that can be used in a systematic fashion to create high-quality air barrier systems. Design details are being developed by CMHC for all wall types to show how air barrier systems must be integrated with wall construction. For example, CMHC *Best Practice Guides* for wall construction are being developed with 4-d design details—the fourth dimension being that of time. Drawings will be provided that show how the building envelope system must be assembled, or layered, to control not only air leakage but also vapor diffusion, heat loss, and rain penetration. Construction

quality control specifications and procedures have also been established by organizations such as the National Air Barrier Association to ensure that the design details are achieved in practice.

With this evolution, one would expect to see the quality of air barrier systems improving. But unfortunately, designers and developers often see air barrier technologies as too expensive to design, install, and test and the results of doing so too intangible; they

have been slow to fully embrace what they typically see as a nuisance. By throwing air barrier material and sealants into the design and construction of buildings, most design teams assume that they are adequately addressing the air leakage control issue. However, without a systematic quality assurance program in place to ensure the proper application, installation, and testing of all the components of an air barrier system, most building owners can only assume that an air barrier has been put in place. Postconstruction experience often shows that this is not the case. What's worse is that the remedial costs are seen as being so prohibitive that the problem is never fully addressed.

It's not all bad news, though. Several recent construction projects have demonstrated that where there's a will (and a budget), there's a way. A 48-unit, six-story apartment building con-

structed recently in Dundas, Ontario, (photo on p. 29) achieved an air leakage rate of 1.18 L/s/m² at 75 Pa—more than 10 times higher than recommended, and the best result achieved in a high-rise multifamily building to date in Canada. The building energy and environment design consultant, Enermodal Engineering of Waterloo, Ontario, developed an air leakage control program that:

- oversaw the development of appropriate air barrier system design details;
- initiated the construction trades into the principles of air barrier system installation;
- provided periodic review of air barrier details;
- tested the air barrier system as it was installed; and
- certified that the original performance target for air leakage was achieved.

Postconstruction monitoring of the building found it to be very energy efficient (50% overall energy reduction compared to conventional apartment buildings). It was also found to be well ventilated and comfortable because balanced heat recovery systems were installed in each apartment.

A similar air leakage control program by the Building Performance Section of the Saskatchewan Research Council showed even better results in the design and construction of a library building and a small regional health building. The projects resulted in air leakage rates of 0.23 L/s/m² at 75 Pa and 0.19 L/s/m² at 75 Pa, respectively. So it's possible that apartment buildings could be built equally airtight.

Table 1. Air Leakage Characteristics of Multifamily Buildings

Data Type *	Mean (L/s/m ² at 75 Pa)	Range (L/s/m ² at 75 Pa)	Standard Deviation (L/s/m ² at 75 Pa)
Type 1 (12 buildings)	3.19	1.18–6.37	1.24
Type 2 (3 buildings)	4	3.15–4.5	0.6
Type 3 (8 buildings)	4.3	0.83–10	2.73

* Type 1 is for whole-building tests. Type 2 is for building tests where the result was reported as a function of the exterior wall area only. Type 3 covers data generated by floor-by-floor or single-suite tests.

Pushing the Benefits

While these results are encouraging, there is still much to do before the integration of air leakage control becomes common in the design, construction, and commissioning of air barrier systems. In the absence of specific building code regulations, tools must be developed to convince developers and building owners that air leakage control makes sound economic sense. Developers, who have no vested interest in the long-term operational energy and cost savings attributable to air leakage control, have to be shown that an air leakage control program will improve the overall quality of the building envelope, thereby limiting potential construction delays, wall defects, warranty claims, and liability. (More work is required to identify the specific costs associated with a thorough air leakage control program so that a comparison with the benefits becomes possible.) Building owners need to be shown that long-term operational costs associated with space conditioning and building envelope maintenance and repair can be reduced. Without tangible evidence of the benefits, air leakage control will continue to be a difficult commodity to sell at the design stage.

CMHC is currently exploring ways to develop tools to take some of the mystery out of integrating air leakage control in large buildings. Calculation methods for estimating the potential benefits of meeting specific air barrier leakage targets are being explored for new construction and retrofit opportunities. Airtightness test methods and equipment for large-building applications must also be developed in coordination with the suppliers of more conventional blower door devices. CMHC is also continuing to support the training of architects and other building professionals through institutions such as the AIA and the Royal Architectural Institute of Canada on the design of air barrier systems. Such activities may help to make air leakage control a common practice before it becomes necessary to mandate it in building codes and standards.



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